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// 📝 Mock Practical Exam Paper - Variant 3 (90 Minutes)
// 🔍 Focus: Missed or Less-Used Concepts from Previous Papers

// ✅ Question 1 - Static Fields & Methods

public abstract class HotelStaff {
    private static int staffCount = 0;

    public HotelStaff(String name, String surname) {
        this.name = name;
        this.surname = surname;
        staffCount++;
    }

    public static int getStaffCount() {
        return staffCount;
    }
    // ... existing fields/methods
}

// ✅ Question 2 - Logger Class Integration

import java.util.logging.*;

public class WestminsterHotelManager implements HotelManager {
    private static final Logger logger = Logger.getLogger("HotelLogger");

    public void addStaffToList(HotelStaff staff) {
        if (hotelStaffList.size() < staffLimit) {
            hotelStaffList.add(staff);
            logger.log(Level.INFO, staff.getClass().getSimpleName() + " " +
staff.getName() + " added.");
        }
    }
}

// ✅ Question 3 - Abstract Method in HotelStaff

public abstract class HotelStaff {
    public abstract String getRoleDescription();
}

public class Manager extends HotelStaff {
    @Override
    public String getRoleDescription() {
        return "Manages hotel operations";
    }
}

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public class HouseKeeper extends HotelStaff {
    @Override
    public String getRoleDescription() {
        return "Responsible for cleaning rooms";
    }
}

// ✓ Question 4 - Queue for Guest Check-in

Queue<Guest> guestQueue = new LinkedList<>();

public void processGuestQueue() {
    while (!guestQueue.isEmpty()) {
        Guest g = guestQueue.poll();
        System.out.println("Checked in: " + g.getName() + " " + g.getSurname());
    }
}

// ✓ Question 5 - Comparator by Age

public class StaffAgeComparator implements Comparator<HotelStaff> {
    public int compare(HotelStaff a, HotelStaff b) {
        return a.getDateOfBirth().compareTo(b.getDateOfBirth());
    }
}

public void printSortedStaffByAge() {
    hotelStaffList.sort(new StaffAgeComparator());
    for (HotelStaff s : hotelStaffList) {
        System.out.println(s);
    }
}

// ✓ Question 6 - Duplicate Staff ID Exception

public class DuplicateStaffIDException extends Exception {
    public DuplicateStaffIDException(String message) {
        super(message);
    }
}

// Inside addHotelStaff()
for (HotelStaff s : hotelStaffList) {
    if (s.getStaffID().equals(staffID)) {
        throw new DuplicateStaffIDException("Staff ID already exists!");
    }
}

// ✓ Question 7 - Convert to Array

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public void printGuestArray() {
    Guest[] array = hotelGuestList.toArray(new Guest[0]);
    for (Guest g : array) {
        System.out.println(g.getName() + " " + g.getSurname());
    }
}

// ✓ Question 8 - GUI JComboBox Filter

JComboBox<String> roleFilter = new JComboBox<>(new String[]{"All", "Manager",
"HouseKeeper"});
roleFilter.addActionListener(e -> {
String role = (String) roleFilter.getSelectedItem();
ArrayList<HotelStaff> filtered = new ArrayList<>();
    for (HotelStaff s : list) {
        if (role.equals("All") ||
            (role.equals("Manager") && s instanceof Manager) ||
            (role.equals("HouseKeeper") && s instanceof HouseKeeper)) {
            filtered.add(s);
        }
    }
    tableModel.updateList(filtered); // custom method to update
table list
    tableModel.fireTableDataChanged();
});
add(roleFilter, BorderLayout.NORTH);

// ✓ Question 9 - Countdown Exit Thread

public class ExitCountdownThread extends Thread {
    public void run() {
        try {
            for (int i = 5; i > 0; i--) {
                System.out.println("Exiting in " + i + "...");
                Thread.sleep(1000);
            }
            System.out.println("Timer ended - auto closing system");
            System.exit(0);
        } catch (InterruptedException e) {
            e.printStackTrace();
        }
    }
}

// Call this in runMenu() when case 0 is triggered
new ExitCountdownThread().start();

// ✓ End of Variant 3 Solutions

```